

Exercise 1 – HTML & Tailwind

Weighting: 2%

For this week's exercise, we ask all students to use our development environment Elf. If you prefer to use your own development environment, you can do so from next week.

In this exercise we are going to create simple HTML and Tailwind web pages.

Task 1 – Development Environment and simple HTML page.

Follow the lecture to create a simple web page on Elf.

1. On your Elf machine, create directories `~/html/webAppDev/week1/task1`, and inside this directory, create a file `index.html`. Place the following simple HTML code in this file:

```
<!DOCTYPE html>
<html>
<head>
  <meta charset=utf8>
  <title>Hello World!</title>
</head>
<body>
  <h1>Hello World!</h1>
</body>
</html>
```

View this html file in a web browser.

2. Zip up the *task1* project/directory and download it from Elf.
3. Set a password to restrict access to all your websites via `.htaccess` file.

Task 2 – Learn Tailwind with AI as Your Frontend Tutor

Place any code you write for this task in `~/html/webAppDev/week1/task2` directory.

Being able to learn from AI has become an important skill for software developer. Your task is to learn the **Tailwind CSS** framework using AI.

1. Open your preferred AI tool (ChatGPT, Claude, Copilot).
2. **Prompt the AI:** *"I am a backend developer who only knows basic HTML. Explain what Tailwind CSS is, how it differs from traditional CSS, and give me the HTML `<script>` tag to use Tailwind via CDN for a quick prototype."*
3. **Prompt for Basics:** Ask the AI: *"Explain how Tailwind handles margins, padding, text colors, and the Flexbox grid. Give me a 5-minute crash course with small code examples."*
4. Read the AI's response. You must understand how utility classes work (e.g., that `mt-4` means margin-top, `text-center` centers text, etc.). Copy the code example into a html file, and view the webpage on a browser. Ask AI to explain any part of the code example you do not understand.

Task 3 – Generate and Audit a Responsive Layout

Place any code you write for this task in `~/html/webAppDev/week1/task3` directory.

Now, use AI to generate your page layouts. You will create two separate HTML files: `index.html` (the feed) and `profile.html` (the user profile).

1. **The Main Page (`index.html`):** Ask the AI to generate a responsive HTML page for a Social Media feed using Tailwind CSS. It must include a Navigation bar, a "New Post" form, and a list of at least 3 posts. It must be a 2-column layout on desktop, and collapse to a 1-column layout on mobile devices.
2. **Carefully study the code from AI.** If there is any part of the code you do not understand, you can type (copy the code) into AI chat and ask AI to explain it.
3. **The Profile Page (`profile.html`):** In the same AI chat (so it remembers your design), prompt the AI: *"Now generate a User Profile page (`profile.html`) using the same Navbar and styling. It should display a profile picture, a short bio, and a grid of the user's photos."*
4. **The Connection:** You must ensure the two pages are linked. The Navbar on `index.html` must have a link that opens `profile.html`, and `profile.html` must have a "Home" link that brings you back to `index.html`.
5. **The Audit (Your Job):** Paste the code into your files and view them in the browser. Look at the HTML tags. Find the Tailwind classes that handle the mobile breakpoint (these usually look like `flex-col md:flex-row` or `w-full md:w-1/2`).

Task 4 – The Manual Tweaks (Proving you learned it)

AI code is never exactly what you want. You must prove you learned how to read Tailwind by making manual changes **without** asking the AI to write the code for you.

Part A: Mandatory Tweaks

Apply these three changes manually to your HTML:

1. Change the background color of the Navigation bar on **both pages** to a different color (e.g., change `bg-blue-600` to `bg-emerald-800`).
2. Add rounded corners to the images in your posts and profile picture (hint: look up the `rounded-full` or `rounded-lg` utility classes).
3. Add a drop-shadow to the "New Post" form box.

Part B: AI-Guided Exploration (Self-Study)

To ensure you are fully prepared for the "Live Mutation" test in your peer review, you should practice making more tweaks. Use AI to generate practice tasks for yourself.

Important: Do this in the *exact same chat session* you used for Task 3. This ensures the AI already knows what your HTML code looks like.

Copy and paste this exact prompt into your AI tool:

"Based on the HTML you just generated for my social media feed, I want to practice modifying Tailwind classes. Act as my tutor. Give me 3 small, specific design tweaks I can manually apply to my HTML to improve the design (for example: adding hover effects to the submit button, adjusting the spacing between posts, or changing text colors)."

CRITICAL RULE: Do NOT write the exact code for me. Only tell me what visual effect to achieve, and tell me the general names of the Tailwind utility categories I should look up in the documentation to figure it out myself."

IN-CLASS PEER REVIEW PROTOCOL

During the lab class, students will review each other in small groups. The core purpose of this peer review is to practice explaining code and proving that you understand the AI's output.

As the Reviewer:

Your group must take turns so that each student asks exactly one live mutation question and performs exactly one live mutation test.

To test your peer's understanding of Tailwind and basic HTML, ask your reviewee ONE question to manually change their Task 4 code. Your question should be similar in nature, but different to the following examples:

- "Show me your HTML file. I see the text here is text-lg. Change it to be significantly larger right now."
- "Change the background color of the 'Submit' button from blue to red."
- "Show me the exact <a> tag that links your feed to your profile page. Now, change the link text from 'Profile' to 'My Account'."

(Note: You can ask AI to generate the mutation question for you to ask).

As the Reviewee:

Your primary goal is to demonstrate your application and explain *how* you manually modified the AI-generated code. You must do the following:

1. **Demonstrate:** Show your two working pages in the browser. Click back and forth to prove the Feed and Profile pages are properly linked, and resize the browser to show the responsive layout working.
2. **Explain your Code (Task 4 Focus):** Open your HTML files and explain to your reviewer exactly how you performed the manual tweaks. You must point to the code and explain:

- **The Navbar:** Where the Navbar code is located, and the exact Tailwind class you changed to update the background color across *both* files.
 - **The Images:** The specific utility class you used to add rounded corners (e.g., explaining that rounded-full makes a circle, or rounded-lg makes soft corners).
 - **The Form:** Where the "New Post" HTML block begins, and the class you added to create the drop-shadow effect.
3. **Perform the Live Mutation:** Successfully complete the on-the-fly change requested by your peer reviewer without using AI.
 4. **Show your Process:** Briefly show your prompt history to prove you asked the AI to *teach* you the framework, rather than just asking it to write the code.